

IEEE Innovation Nation Sri Lanka 2026

Zonal Competitions

Sample Proposal



Byte Busters

Zone D

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1. Team Overview

1.1 Team Name: Byte Busters

1.2 Team Members:

P. D. D. S. Kulasekara (Hardware & Embedded Systems), H. H. A. M. Sethumika (Mobile Application Development), W. R. D. I. Sandeepani (Web Dashboard & UI/UX), C. H. Amarasinghe (Cloud Backend & System Integration), T.S Perera (IoT Sensor Testing)

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2. Problem Statement

Forest fires are among the most destructive natural disasters globally, causing widespread ecological damage, loss of wildlife, and significant economic harm. As global warming intensifies and dry seasons grow longer, the frequency and severity of wildfires are increasing at an alarming rate across the world, including Sri Lanka.

2.1 What is the problem?

Existing detection methods — including manual patrols, satellite surveillance, and stationary camera systems — are slow to detect early-stage ignitions, lack ground-level coverage in dense forest terrain, are expensive to maintain, and offer no intelligent risk assessment. By the time a fire is identified, it may have already spread to a dangerous scale. There is no single, affordable, autonomous, and intelligent ground-level system capable of continuous off-grid forest monitoring in Sri Lanka.

2.2 Who experiences this problem?

Government forest departments and environmental authorities, disaster management units, wildlife conservation organisations, hikers and local communities near forested areas, and national economies impacted by ecological and agricultural losses from wildfires.

2.3 Why is this problem important?

Delayed detection leads to rapid fire spread, irreversible ecological damage, and substantial loss of life and property. Sri Lanka's forested areas cover nearly 30% of the land, many lacking any automated early-warning system. A reliable, autonomous, and continuously operating fire detection platform has the potential to dramatically reduce wildfire impacts on the environment, economy, and communities.

3. Proposed Solution

The Smart Forest Guardian is a solar-powered, autonomous IoT robotic platform designed to continuously patrol forest environments and detect fire risks in real time, without requiring grid power or human intervention.

3.1 What is your solution?

An autonomous, solar-powered robotic platform equipped with a multi-sensor array (MLX90640 thermal camera, MQ-9 gas sensor, DHT11 temperature and humidity sensor, rain sensor), an ESP32 Wi-Fi microcontroller, and a cloud-connected Severity Level Classification System that continuously monitors forest environments and delivers real-time fire risk alerts via a mobile app and web dashboard.

3.2 How does it work?

The robot autonomously patrols forest terrain. The ESP32 microcontroller reads all sensor data including thermal imaging, gas concentrations, temperature, humidity, and rainfall and transmits it via Wi-Fi to a Firebase Realtime Database. Sensor data is analysed in the cloud to classify fire risk into Low, Medium, or High severity levels and triggers push notifications to the React Native mobile app and React.js web dashboard. Two user roles are supported: Government Officers (full system access with sensor feeds and alerts) and Hikers (safety dashboard with fire warnings and emergency tips). The solar panel and LiPo battery ensure continuous off-grid operation.

3.3 Unique Value Proposition

Unlike existing systems, the Smart Forest Guardian combines four capabilities into one platform: Thermal imaging + multi-sensor fusion for highly accurate, low-false-alarm detection even in smoke-obscured conditions; Autonomous ground-level coverage in dense terrain where aerial or satellite systems fail; Solar-powered sustainability for indefinite off-grid operation; Low hardware cost (approx. Rs. 27,900 LKR) enabling large-scale institutional deployment.

4. Market Opportunity

The Smart Forest Guardian targets government and institutional bodies responsible for forest conservation and disaster management, both in Sri Lanka and across the South and Southeast Asian region.

4.1 Target Market

Sri Lanka Department of Forest Conservation, Disaster Management Centre

(DMC), national park authorities, wildlife conservation organizations, eco-tourism operators, and international NGOs involved in forest protection.

4.2 Market Size

Sri Lanka alone has over 900,000 hectares of government-protected forest reserves, most lacking automated fire monitoring infrastructure. The global wildfire detection and monitoring market is growing rapidly due to increasing wildfire incidents worldwide. Regionally, Southeast Asian nations face similar challenges, representing a significant addressable market for scalable, affordable autonomous fire detection.

4.3 Market Need

Current government systems rely predominantly on manual patrols or expensive satellite subscriptions that provide delayed, low-resolution data. There is a clear and urgent need for a ground-level, real-time, self-sustaining solution deployable at scale with minimal operational cost exactly what the Smart Forest Guardian delivers.

5. Business Model

The Smart Forest Guardian generates revenue through hardware sales, software subscriptions, and institutional contracts.

5.1 Revenue Streams

Hardware unit sales to government departments and national parks; SaaS subscription for the cloud monitoring dashboard and alert management platform; annual maintenance and support contracts; government and institutional tender projects for large-scale forest monitoring deployments.

5.2 Pricing Strategy

Hardware units priced competitively based on the approx. Rs. 28,500 LKR production cost per unit. SaaS dashboard offered on a tiered subscription model (basic for small reserves, premium for multi-device deployments). Government procurement approached through bulk contract pricing.

5.3 Cost Structure (Optional)

Hardware cost per unit: ESP32 (Rs. 3,500) + MLX90640 Thermal Camera (Rs. 5,500) + MQ-9 Gas Sensor (Rs. 1,500) + DHT11 Sensor (Rs. 500) + Rain Sensor (Rs. 700) + Servo Motor (Rs. 1,200) + Solar Panel & Battery (Rs. 6,000) + Robot Chassis (Rs. 4,000) + ESP32-CAM (Rs. 4,000) + Wires & PCB (Rs. 1,000) = Total Rs. 27,900 LKR.

6. Progress and Validation (If Applicable)

The system has been validated through prototype development, simulation testing, and stakeholder interviews with forest monitoring personnel.

6.1 Users / Customers

Prototype tested with simulated fire scenarios. Two user roles validated: Government Officers and Hikers. Stakeholder interviews conducted with forest monitoring personnel to shape system requirements and UI design.

6.2 Validation

Environmental simulation testing confirmed Low, Medium, and High severity classification accuracy. Unit and integration testing completed across ESP32 firmware, Firebase cloud sync, thermal camera feed, and mobile push notifications. UAT conducted on web dashboard and mobile app. Thermal camera validated for detecting heat anomalies in smoke-obscured conditions where optical cameras fail.

6.3 Key Metrics

All three severity levels (Low / Medium / High) confirmed accurate in simulation; sensor data transmission latency under 2 seconds via Firebase; thermal imaging detection range validated at 5–7 metres per unit; solar panel sustains continuous operation without grid dependency; false alarm rate significantly reduced through multi-sensor fusion.

7. Future Plans

The Smart Forest Guardian is envisioned as a nationally deployable wildfire prevention infrastructure for Sri Lanka, with the potential for regional expansion across South and Southeast Asia.

7.1 Short-Term Goals

Conduct real-world pilot deployment in a controlled Sri Lankan forest reserve; enhance thermal camera hotspot detection algorithms; expand the cloud dashboard with historical trend analysis and incident reports; complete full-scale UAT with forest officers and disaster management personnel.

7.2 Long-Term Vision

Multi-unit swarm deployment covering large forest tracts; integration with national disaster management platforms for automated emergency dispatch; incorporation of machine learning algorithms to enable predictive fire risk forecasting based on accumulated sensor data and historical patterns; government and international institutional partnerships for grant-funded large-scale deployment; open API ecosystem for third-party conservation tools.

8. Additional Information (Optional)

The Smart Forest Guardian addresses key gaps in current forest fire detection: ground-level thermal imaging, multi-sensor fusion, autonomous off-grid operation, and real-time multi-role alerts all in a single cost-effective platform.

8.1 Risks and Challenges

Wi-Fi connectivity limitations in deep forest terrain (mitigation: LoRa/mesh networking as future upgrade); reduced solar energy under heavy canopy (mitigation: LiPo battery buffering and ESP32 low-power sleep cycles); thermal camera accuracy in high-humidity conditions (mitigation: multi-sensor fusion with gas and environmental sensors); rough terrain navigation requiring robust chassis design.

8.2 Competitive Analysis

Satellite systems: delayed data, no ground coverage, high cost. Stationary cameras: fixed position, no thermal imaging, no automated risk classification. Manual patrols: slow, high human cost, no real-time alerts. Smart Forest Guardian advantage: ground-level thermal imaging + multi-sensor fusion + automated severity classification + autonomous solar-powered operation + real-time alerts, all at a low unit cost of approx. Rs. 27,900 LKR.

8.3 Expansion Plans

Phase 1: Pilot in one Sri Lankan forest reserve (3–5 units). Phase 2: Scale to all major government-managed reserves across Sri Lanka. Phase 3: Regional expansion to South and Southeast Asia. Phase 4: Integration with international wildfire response networks and licensing of the Smart Forest Guardian platform to foreign environmental agencies.